Text Large Model + robotic arm control（Voice Version）

Before running the function, you need to close the App and large programs. For the closing method, refer to [4. Preparation] - [1. Manage APP control services].

1. Function Description

After the program runs, wake up the voice module and say a sequence of robotic arm control commands. The large model will plan the action functions corresponding to the action commands and execute all commands in sequence.

2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py
```

Currently available robotic arm control commands include:

- **Individual servo control**: Set servo x to y degrees, where x ranges from 1-6 and y ranges from 0-180, for example, set servo 1 to 120 degrees.
- **Individual robotic arm action control**: Robotic arm up/down/left/right, gripping posture, tracking posture, for example, set the robotic arm to gripping posture, robotic arm stretch upward;
- **Control robotic arm motion sequences**: Robotic arm dance/clap/nod/shake head, for example, please nod your head;
- **Change robotic arm end position**: Robotic arm adjust up/down/left/right/forward/backward by x cm, for example, move the robotic arm up by 3 cm

After waking up the module, give robotic arm control commands. You can refer to the following example:

```
First, adjust the robotic arm upward by 3 centimeters, wait 3 seconds, then
extend the robotic arm downward, stay for 3 seconds, then have the robotic arm
shake its head, pause for 3 seconds, and finally set servo 1 to 150 degrees.
```

```
[asr-5] [INFO] [1775200637.708199902] [asr]: -
[asr-5] [INFO] [1775200637.768089953] [asr]: -
[asr-5] [INFO] [1775200637.828238155] [asr]: -
[asr-5] [INFO] [1775200637.888032835] [asr]: -
[asr-5] [INFO] [1775200641.939551637] [asr]: XunFei ASR Finish.
[asr-5] [INFO] [1775200641.940254604] [asr]:  First adjust the robotic arm upward by three centimeters, wait three seconds, then extend the robotic arm down
ward stay for three seconds, then have the robotic arm, shake its head, pause for three seconds, and finally set, serve one to 150 degrees.
[asr-5] [INFO] [1775200641.941318332] [asr]: 😐okay, let me think for a moment...
[model_service-3] [INFO] [1775200658.304371249] [model_service]: Decision making AI planning:Perform the following steps in sequence:
[model_service-3]
[model_service-3] 1. Call the function to move the robotic arm upward by three centimeters.
[model_service-3] 2. Call the function to wait for three seconds.
[model_service-3] 3. Call the function to move the robotic arm downward by three centimeters.
[model_service-3] 4. Call the function to wait for three seconds.
[model_service-3] 5. Call the function for the robotic arm to shake its head.
[model_service-3] 6. Call the function to wait for three seconds.
[model_service-3] 7. Call the function to adjust joint 6 to 150 degrees.
[model_service-3] [INFO] [1775200660.384421280] [model_service]: "json_str": {"response": "Got it! I'll follow your instructions step by step, like a well-c
horeographed dance. Let's begin!", "action": ["arm_move('up', 3)"]}
[model_service-3] [INFO] [1775200660.385155081] [model_service]: "action": ["arm_move('up', 3)"], "response": Got it! I'll follow your instructions step by
step, like a well-choreographed dance. Let's begin!
[action_service-4] [INFO] [1775200660.392162635] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200660.392831194] [action_service]: "actions": ["arm_move('up', 3)"]
[action_service-4] [INFO] [1775200668.851051255] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.855143961] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.859182242] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.865376218] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.869467739] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.873896466] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.874478728] [action_service]: Dir: up
[action_service-4] [INFO] [1775200668.875134343] [action_service]: Dist: 3
[action_service-4] [INFO] [1775200668.876008662] [action_service]: cur_joints: [0, 0, 0, 0, 0, 0]
[kinematics_dofbot-1] [WARN] [1775200668.938218170] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[kinematics_dofbot-1] [INFO] [1775200668.939471602] [rclcpp]: fk sent***
[action_service-4] [INFO] [1775200670.878584568] [action_service]: CurEndPos: [0.0020449992274857477, -0.06338999792660725, 0.040402443637541426, 1.57080366
10372729, 3.650393812038131e-08, -1.5707926657577578]
[action_service-4] [INFO] [1775200670.879199830] [action_service]: request: dofbot_interface.srv.Kinematics_Request(tar_x=0.0020449992274857477, tar_y=-0.06
338999792660725, tar_z=0.07040244363754142, roll=1.5708036610372729, pitch=3.650393812038131e-08, yaw=-1.538546913711967, cur_joint1=0.0, cur_joint2=0.0, cu
r_joint3=0.0, cur_joint4=0.0, cur_joint5=0.0, cur_joint6=0.0, kin_name='ik')
[kinematics_dofbot-1] [WARN] [1775200670.882166179] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[action_service-4] [INFO] [1775200672.387951878] [action_service]: joints: [0.0, 163, 195, 182, 90, 30]
[action_service-4] [INFO] [1775200674.389933632] [action_service]: Moving.
[action_service-4] [INFO] [1775200674.390737692] [action_service]: self.combination_mode: False
[action_service-4] [INFO] [1775200674.391390936] [action_service]: self.interrupt_flag: False
[action_service-4] [INFO] [1775200674.392014143] [action_service]: Move done.
[action_service-4] [INFO] [1775200674.392819462] [action_service]: Published message: Robot feedback: Execution arm_move_done() completed
[model_service-3] [INFO] [1775200676.547303377] [model_service]: "json_str": {"response": "Alright, I've moved upward by 3 centimeters. Now, let's take a li
ttle pause—just like taking a breath before the next move.", "action": ["wait(3)"]}
[model_service-3] [INFO] [1775200676.547890991] [model_service]: "action": ['wait(3)'], "response": Alright, I've moved upward by 3 centimeters. Now, let's
take a little pause—just like taking a breath before the next move.
[action_service-4] [INFO] [1775200676.552125883] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200676.552642090] [action_service]: "actions": ['wait(3)']
[action_service-4] [INFO] [1775200690.343919110] [action_service]: Published message: Robot feedback: Execute wait(3.0) completed
[model_service-3] [INFO] [1775200693.699834863] [model_service]: "json_str": {"response": "Time's up! Now, I'm going to move downward by 3 centimeters—smoot
h and steady, just like a slow descent.", "action": ["arm_move('down', 3)"]}
[model_service-3] [INFO] [1775200693.701079332] [model_service]: "action": ["arm_move('down', 3)"], "response": Time's up! Now, I'm going to move downward b
y 3 centimeters—smooth and steady, just like a slow descent.
[action_service-4] [INFO] [1775200693.707094030] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200693.707823404] [action_service]: "actions": ["arm_move('down', 3)"]
[action_service-4] [INFO] [1775200702.888654190] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.893407678] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.897793997] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.901859240] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.906301244] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.910175598] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.910802453] [action_service]: Dir: down
[action_service-4] [INFO] [1775200702.911537698] [action_service]: Dist: 3
[action_service-4] [INFO] [1775200702.912177719] [action_service]: cur_joints: [0, 0, 0, 0, 0, 0]
[kinematics_dofbot-1] [WARN] [1775200702.918310881] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[kinematics_dofbot-1] [INFO] [1775200702.918673679] [rclcpp]: fk sent***
[action_service-4] [INFO] [1775200704.915148529] [action_service]: CurEndPos: [0.0020449992274857477, -0.06338999792660725, 0.040402443637541426, 1.57080366
10372729, 3.650393812038131e-08, -1.5707926657577578]
[action_service-4] [INFO] [1775200704.915820681] [action_service]: request: dofbot_interface.srv.Kinematics_Request(tar_x=0.0020449992274857477, tar_y=-0.06
338999792660725, tar_z=0.010402443637541427, roll=1.5708036610372729, pitch=3.650393812038131e-08, yaw=-1.538546913711967, cur_joint1=0.0, cur_joint2=0.0, c
ur_joint3=0.0, cur_joint4=0.0, cur_joint5=0.0, cur_joint6=0.0, kin_name='ik')
[kinematics_dofbot-1] [WARN] [1775200704.919417126] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[action_service-4] [INFO] [1775200706.424566003] [action_service]: joints: [0.0, -86, 155, 291, 90, 30]
[action_service-4] [INFO] [1775200708.427442979] [action_service]: Moving.
[action_service-4] [INFO] [1775200708.431167720] [action_service]: self.combination_mode: False
[action_service-4] [INFO] [1775200708.431694927] [action_service]: self.interrupt_flag: False
[action_service-4] [INFO] [1775200708.432627450] [action_service]: Move done.
[action_service-4] [INFO] [1775200708.433415899] [action_service]: Published message: Robot feedback: Execution arm_move_done() completed
[model_service-3] [INFO] [1775200710.524737907] [model_service]: "json_str": {"response": "Downward movement complete! Now, I'll take another brief pause—th
ree seconds of calm before the next action.", "action": ["wait(3)"]}
[model_service-3] [INFO] [1775200710.525396558] [model_service]: "action": ['wait(3)'], "response": Downward movement complete! Now, I'll take another brief
 pause—three seconds of calm before the next action.
[action_service-4] [INFO] [1775200710.534062270] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200710.534598921] [action_service]: "actions": ['wait(3)']
[action_service-4] [INFO] [1775200722.187367767] [action_service]: Published message: Robot feedback: Execute wait(3.0) completed
[model_service-3] [INFO] [1775200724.764601885] [model_service]: "json_str": {"response": "Alright, time to shake things up a bit—literally! I'm going to sh
ake my head now, just like a robot doing its best 'no' dance.", "action": ["arm_shake()"]}
[model_service-3] [INFO] [1775200724.765178962] [model_service]: "action": ['arm_shake()'], "response": Alright, time to shake things up a bit—literally! I'
m going to shake my head now, just like a robot doing its best 'no' dance.
[action_service-4] [INFO] [1775200724.769843097] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200724.770289488] [action_service]: "actions": ['arm_shake()']
[action_service-4] [INFO] [1775200742.607277555] [action_service]: Published message: Robot feedback: Execute arm_shake() completed
[model_service-3] [INFO] [1775200746.450718174] [model_service]: "json_str": {"response": "Head shake complete! Now, I'm going to adjust my joint 6 to 150 d
egrees—precision at its finest!", "action": ["adjust_joint(6, 150)"]}
[model_service-3] [INFO] [1775200746.451214399] [model_service]: "action": ['adjust_joint(6, 150)'], "response": Head shake complete! Now, I'm going to adju
st my joint 6 to 150 degrees—precision at its finest!
[action_service-4] [INFO] [1775200746.455631885] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200746.456283295] [action_service]: "actions": ['adjust_joint(6, 150)']
[action_service-4] [INFO] [1775200758.768324659] [action_service]: Published message: Robot feedback: Execution adjust_joint_done() completed
[model_service-3] [INFO] [1775200761.954300553] [model_service]: "json_str": {"response": "All done! I've completed every step—up, pause, down, pause, shake
, pause, and finally adjusted joint 6 to 150 degrees. Mission accomplished!", "action": ["finishtask()"]}
[model_service-3] [INFO] [1775200761.955190428] [model_service]: "action": ['finishtask()'], "response": All done! I've completed every step—up, pause, down
, pause, shake, pause, and finally adjusted joint 6 to 150 degrees. Mission accomplished!
[action_service-4] [INFO] [1775200761.960818642] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200761.962118205] [action_service]: "actions": ['finishtask()']
```

3. Core Code Analysis

You can refer to the content in **3. Core Code Analysis** from tutorial [18. AI Model - Text Version] - [Text Large Model + robotic arm control]. The voice version and text version have the same action functions, only the task command input method is different.